1. OVERVIEW & TOP HARDWARE CONTROLS

Launchpad X hardware layout, CC functions, and SysEx mode architecture

HARDWARE INTEGRATION ARCHITECTURE

* Hardware: Novation Launchpad X USB MIDI Controller in bidirectional Programmer / DAW Mode.
* Dual Ports: Live Port 1 (Standard MIDI) and DAW Port 2 (SysEx Handshake & LED RGB Matrix).
* Web MIDI & Standalone: Works seamlessly in Chrome/Edge Web MIDI and headlessly on Raspberry Pi Zero W.
* Zero Feedback Loop: Mode switching uses physical velocity filters (vel=127) and debounced SysEx commands.

TOP ROW CONTROL BUTTONS (CC 91 - CC 98)

CC 91 **UP ARROW** — LED: White glow

Octave Up / Tempo Up. In Step Sequencer mode, shifts octave of melodic tracks up by +12 semitones or increments BPM by +2.

CC 92 **DOWN ARROW** — LED: White glow

Octave Down / Tempo Down. Shifts active melodic track pitch down by -12 semitones or decrements BPM by -2.

CC 93 **LEFT ARROW** — LED: Bright Cyan (Active) / Dim Cyan

Bank 1 / Page 1 Selector. Selects steps 1 through 8 on the 8x8 pad matrix. Lights up bright cyan when Page 1 is active.

CC 94 **RIGHT ARROW** — LED: Bright Cyan (Active) / Dim Cyan

Bank 2 / Page 2 Selector. Selects steps 9 through 16 on the 8x8 pad matrix. Lights up bright cyan when Page 2 is active.

CC 95 **SESSION / MIXER** — LED: Bright Cyan LED indicator

Sequencer Session Mode (DAW Programmer Mode). Sends SysEx [F0 00 20 29 02 0C 10 01 F7] and [0E 01 / 00 7F]. Lights up the 8x8 step sequencer grid and enables playhead sweep.

CC 96 **NOTE MODE** — LED: Bright Magenta/Purple LED indicator

Standalone Melodic Note Mode. Releases DAW mode via SysEx [F0 00 20 29 02 0C 10 00 F7] and engages Note layout [00 01]. Launchpad native scale engine, aftertouch, and instrument playing take over.

CC 97 **CUSTOM MODE** — LED: Bright Amber LED indicator

Standalone Custom User Mode. Releases DAW mode and selects Custom 1 Layout [00 04]. Allows mapping custom MIDI CC and Note faders configured via Novation Components.

CC 98 **RECORD / PLAY** — LED: Bright Green (Running) / Dim Red (Stopped)

Master Transport Toggle. Starts and stops the polyphonic 16-step sequencer clock, Web Audio synthesizer voices, and external MIDI clock broadcast.

2. RIGHT-HAND SCENE LAUNCH BUTTONS

Authentic hardware scene column functions (Row 8 down to Row 1 / CC 89 - CC 19)

RIGHT SCENE COLUMN ARCHITECTURE

The 8 circular buttons on the right edge of the Novation Launchpad X transmit Control Change (and Note On in Programmer Mode) on CC 89 down to CC 19. In this sequencer, they provide direct hardware control matching the official silk-screened labels on the real device.

VOLUME — ROW 8 (TOP RIGHT) | CC 89 / Note 89 — **ROLE: ACTIVE TRACK CYCLER**

* Action: Pressing VOLUME cycles through tracks 1 to 8 (Kick -> Snare -> Hat -> Open Hat -> Clap -> Bass -> Lead -> Chord). Instantly focuses the selected track in the Step & Track Inspector decks.
* LED Feedback: Dim Cyan standby glow.

PAN — ROW 7 | CC 79 / Note 79 — **ROLE: VIEW MODE SWITCHER**

* Action: Cycles the active visual workspace between: (1) 8-Track Multi-Track Grid, (2) 64-Step Melodic Single Track, and (3) Euclidean Algorithmic Engine.
* LED Feedback: Dim Cyan / Active mode accent glow.

SEND A — ROW 6 | CC 69 / Note 69 — **ROLE: BANK 1 / PAGE 1 (STEPS 1-8)**

* Action: Switches the 8x8 matrix display to Step Bank 1 (steps 1 through 8). Used for quick page switching during live performances.
* LED Feedback: Bright Yellow when Page 1 is active; Dim Yellow when on Page 2.

SEND B — ROW 5 | CC 59 / Note 59 — **ROLE: BANK 2 / PAGE 2 (STEPS 9-16)**

* Action: Switches the 8x8 matrix display to Step Bank 2 (steps 9 through 16). Allows sequencing the second half of 16-step patterns.
* LED Feedback: Bright Amber/Orange when Page 2 is active; Dim Yellow when on Page 1.

STOP CLIP — ROW 4 | CC 49 / Note 49 — **ROLE: PANIC & INSTANT STOP**

* Action: Emergency all-notes-off and transport halt. Instantly cuts audio synthesis, releases MIDI note-ons, and clears any stuck active voices.
* LED Feedback: Bright Amber when sequencer is running (armed to stop); Dark Gray when stopped.

MUTE — ROW 3 | CC 39 / Note 39 — **ROLE: TRACK MUTE TOGGLE**

* Action: Toggles Mute on the currently selected active track. Muted tracks stop producing sound and external MIDI while keeping pattern data intact.
* LED Feedback: Bright Red if active track is muted; Dim Red if any other track is muted; Dark Gray when no tracks are muted.

SOLO — ROW 2 | CC 29 / Note 29 — **ROLE: TRACK SOLO TOGGLE**

* Action: Toggles Solo on the currently selected active track. When one or more tracks are soloed, only soloed tracks are audible.
* LED Feedback: Bright Green if active track is soloed; Dim Green if other tracks are soloed; Dark Gray when no tracks are soloed.

RECORD ARM — ROW 1 (BOTTOM RIGHT) | CC 19 / Note 19 — **ROLE: MASTER TRANSPORT PLAY / STOP**

* Action: Hardware transport play/stop trigger. Starts the 16th note clock, Ableton Link synchronization, and playhead sweep.
* LED Feedback: Bright Red pulse when running; Dim Red when stopped.

3. 8X8 PAD MATRIX, MODES & ROW COLORS

Polyphonic sequencing, melodic modes, and RGB palette customization

THE 8x8 SILICONE PAD MATRIX (NOTES 11 - 88)

*** 8 Tracks x 8 Steps per page: Tracks 1 to 8 correspond to rows 8 down to 1 on physical Launchpad X.**

- Row 8 (Note 81-88): Track 1 - 808 Sub Kick
- Row 7 (Note 71-78): Track 2 - Snare Drum
- Row 6 (Note 61-68): Track 3 - Closed Hi-Hat
- Row 5 (Note 51-58): Track 4 - Open Hi-Hat
- Row 4 (Note 41-48): Track 5 - Clap / Percussion
- Row 3 (Note 31-38): Track 6 - Acid Resonant Bass
- Row 2 (Note 21-28): Track 7 - Synth Lead Melodic
- Row 1 (Note 11-18): Track 8 - Polyphonic Chord Stab

*** Playhead Visual Sweep: High-speed delta ticks flash white on active beats, sweeping left-to-right.**

*** Unused Pads Toggle: Option in UI to switch idle pads between unlit [OFF] and subtle track-tinted [FADED].**

CORE OPERATIONAL MODES

SESSION / SEQUENCER MODE (DAW PROGRAMMER)

Full 8-track polyphonic step sequencer. The Launchpad X hardware is put into Programmer Mode with DAW mode ON. Pads act as step toggles. Top and Scene buttons function as sequencer transport, octave, bank, and mute/solo controls. Real-time LED feedback reflects the audio clock sweep.

NOTE MODE (STANDALONE SCALES & INSTRUMENT)

Puts the Launchpad X into native Standalone Note Mode (DAW mode released via SysEx). The Launchpad X hardware scale engine takes over. You can play melodic scales, chords, and basslines directly with polyphonic aftertouch. Notes trigger the built-in synth engine and external MIDI outputs.

CUSTOM MODE (USER MAPPINGS)

Selects the hardware Custom 1 layout. Useful for integrating with third-party software, DAW controllers, or custom Novation Components templates while running the sequencer.

ROW COLOR CUSTOMIZATION (NOVATION LPX RGB PALETTE)

You can customize the color of each of the 8 sequencer rows independently:

1. Left Row Track Header Swatches: Click the color dot next to any track row (T1..T8) on the Launchpad chassis.
2. Toolbar "Row Colors" Button: Opens a modal with all 8 rows and the full 14-preset Launchpad X palette.
3. Bottom Track Indicator Strip: Click any track badge to select it or pick a new LED color.
4. Track Inspector Deck: Select any track and click any of the 14 Novation Launchpad X palette circles.

* Immediate Hardware Sync: Selecting any color immediately transmits the exact Novation color index byte to the physical controller LEDs, updating both active step pads and faded background tints instantly.

* 14 LPX Presets: Red (#5), Amber (#9), Orange (#60), Yellow (#13), Lime (#17), Green (#21), Mint (#29), Cyan (#37), Sky (#41), Blue (#45), Purple (#49), Magenta (#53), Pink (#57), White (#3).

4. ABLETON LINK, PI ZERO W & QUICK REFERENCE

Wireless tempo sync, embedded headless deployment, and shortcut reference

ABLETON LINK & RASPBERRY PI ZERO W DEPLOYMENT

* Ableton Link: Built-in wireless tempo and phase synchronization across DAWs, synths, and iOS/Android apps.
  - Quantum and Peer tracking: Displays active connected peers and real-time beat phase metronome (1-4).
* Raspberry Pi Zero W Deployment: Perfect for portable, dawless standalone operation.
  - USB OTG Host: Connect Launchpad X directly to Pi Zero W micro-USB port using an OTG adapter.
  - Headless Python/ALSA Sequencer: Export sequencer patterns as Pi JSON configs directly from the UI.
  - Systemd Service: Automatic launch on boot without requiring a monitor, keyboard, or mouse.
* Polymetric & Euclidean Rhythms: Each track supports independent polymetric loop lengths (1 to 16 steps) and Euclidean algorithmic hit distributions (e.g. E(5, 16) for afro-cuban polyrhythms).

QUICK BUTTON & SHORTCUT REFERENCE TABLE

BUTTON / CONTROL	HARDWARE CC / NOTE	PRIMARY ACTION IN SEQUENCER
Up Arrow	`CC 91`	Octave Shift Up (+12) / Tempo Increment (+2 BPM)
Down Arrow	`CC 92`	Octave Shift Down (-12) / Tempo Decrement (-2 BPM)
Left Arrow	`CC 93`	Switch to Step Bank 1 (Steps 1-8)
Right Arrow	`CC 94`	Switch to Step Bank 2 (Steps 9-16)
Session / Mixer	`CC 95`	Engage 8-Track Step Sequencer (DAW Programmer Mode)
Note Mode	`CC 96`	Release DAW mode; play live scales with hardware engine
Custom Mode	`CC 97`	Switch to Launchpad User Custom 1 Layout
Record / Play	`CC 98`	Master Transport Play / Stop toggle
Volume (Scene 8)	`CC 89 / Note 89`	Cycle selected active track (Track 1 to Track 8)
Pan (Scene 7)	`CC 79 / Note 79`	Cycle view mode (Grid / 64-Step Melodic / Euclidean)
Send A (Scene 6)	`CC 69 / Note 69`	Select Page 1 (Steps 1-8)
Send B (Scene 5)	`CC 59 / Note 59`	Select Page 2 (Steps 9-16)
Stop Clip (Scene 4)	`CC 49 / Note 49`	Panic / Silence all voices and stop playback immediately
Mute (Scene 3)	`CC 39 / Note 39`	Toggle Mute on the active track (lights red when active)
Solo (Scene 2)	`CC 29 / Note 29`	Toggle Solo on the active track (lights green when active)
Record Arm (Scene 1)	`CC 19 / Note 19`	Toggle Transport Play/Stop (lights red when running)
8x8 Silicone Pads	`Notes 11-88`	Toggle individual sequence steps on active page
Unused Pads (UI)	`Toolbar Toggle`	Switch idle pad LEDs between unlit [OFF] and [FADED]
Row Colors (UI)	`Toolbar / Swatches`	Choose from 14 official Novation Launchpad X colors

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